

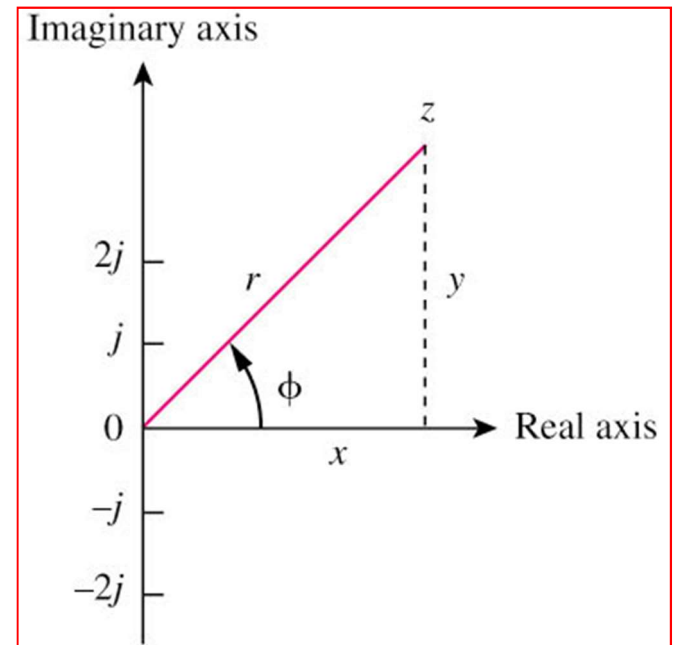
9.3 Phasor (1)

- A phasor is a complex number that represents the amplitude and phase of a sinusoid.
- It can be represented in one of the following three forms:

a. Rectangular $z = x + jy = r(\cos \phi + j \sin \phi)$

b. Polar $z = r \angle \phi$

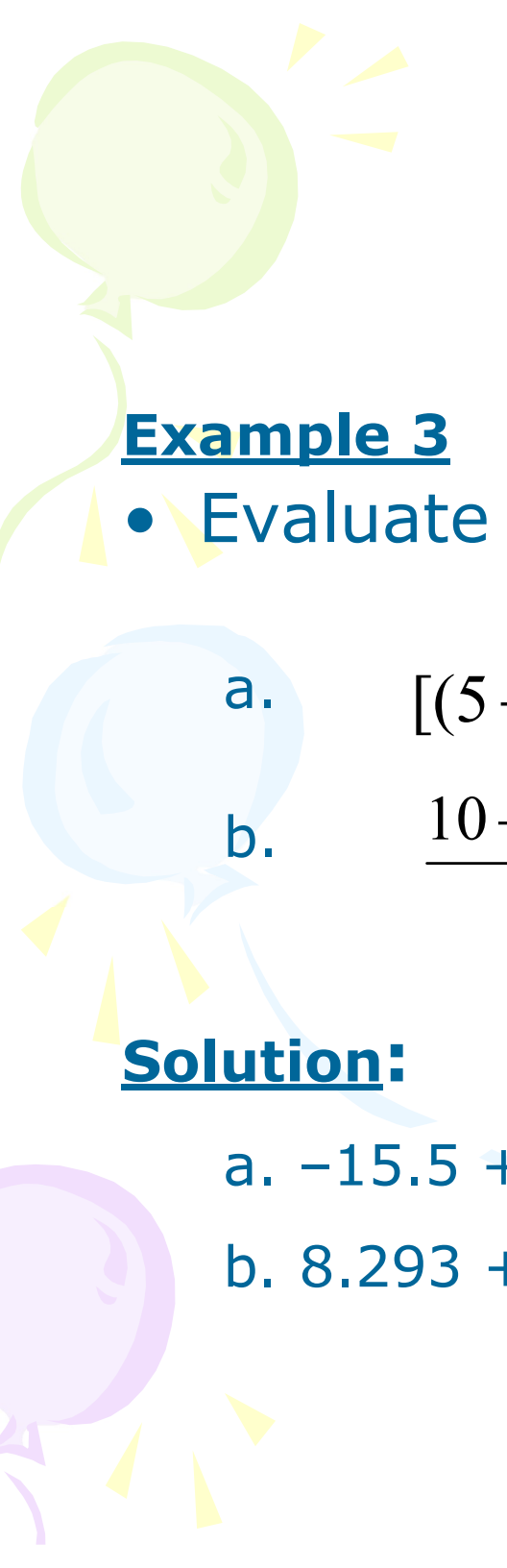
c. Exponential $z = re^{j\phi}$



where

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1} \frac{y}{x}$$



9.3 Phasor (2)

Example 3

- Evaluate the following complex numbers:

a. $[(5 + j2)(-1 + j4) - 5 \angle 60^\circ]$

b. $\frac{10 + j5 + 3 \angle 40^\circ}{-3 + j4} + 10 \angle 30^\circ$

Solution:

a. $-15.5 + j13.67$

b. $8.293 + j2.2$



9.3 Phasor (3)

Mathematic operation of complex number:

1. Addition

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

2. Subtraction

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

3. Multiplication

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$$

4. Division

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$$

5. Reciprocal

$$\frac{1}{z} = \frac{1}{r} \angle -\phi$$

6. Square root

$$\sqrt{z} = \sqrt{r} \angle \phi/2$$

7. Complex conjugate

$$z^* = x - jy = r \angle -\phi = re^{-j\phi}$$

8. Euler's identity

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

9.3 Phasor (4)

- Transform a sinusoid to and from the time domain to the phasor domain:

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow V = V_m \angle \phi$$

(time domain)

(phasor domain)

- Amplitude and phase difference are two principal concerns in the study of voltage and current sinusoids.
- Phasor will be defined from the cosine function in all our proceeding study. If a voltage or current expression is in the form of a sine, it will be changed to a cosine by subtracting from the phase.

9.3 Phasor (5)

Example 4

Transform the following sinusoids to phasors:

$$i = 6\cos(50t - 40^\circ) \text{ A}$$

$$v = -4\sin(30t + 50^\circ) \text{ V}$$

Solution:

a. $I = 6\angle -40^\circ \text{ A}$

b. Since $-\sin(A) = \cos(A+90^\circ)$;

$$v(t) = 4\cos(30t + 50^\circ + 90^\circ) = 4\cos(30t + 140^\circ) \text{ V}$$

Transform to phasor $\Rightarrow V = 4\angle 140^\circ \text{ V}$

9.3 Phasor (6)

Example 5:

Transform the sinusoids corresponding to phasors:

a. $\mathbf{V} = -10\angle 30^\circ \text{ V}$

b. $\mathbf{I} = j(5 - j12) \text{ A}$

Solution:

a) $v(t) = 10\cos(\omega t + 210^\circ) \text{ V}$

b) Since $\mathbf{I} = 12 + j5 = \sqrt{12^2 + 5^2} \angle \tan^{-1}(\frac{5}{12}) = 13\angle 22.62^\circ$
 $i(t) = 13\cos(\omega t + 22.62^\circ) \text{ A}$

Find the sinusoids represented by these phasors:

(a) $\mathbf{I} = -3 + j4 \text{ A}$

(b) $\mathbf{V} = j8e^{-j20^\circ} \text{ V}$

Solution:

(a) $\mathbf{I} = -3 + j4 = 5 \angle 126.87^\circ$. Transforming this to the time domain gives

$$i(t) = 5 \cos(\omega t + 126.87^\circ) \text{ A}$$

(b) Since $j = 1 \angle 90^\circ$,

$$\begin{aligned} \mathbf{V} &= j8 \angle -20^\circ = (1 \angle 90^\circ)(8 \angle -20^\circ) \\ &= 8 \angle 90^\circ - 20^\circ = 8 \angle 70^\circ \text{ V} \end{aligned}$$

Converting this to the time domain gives

$$v(t) = 8 \cos(\omega t + 70^\circ) \text{ V}$$

9.3 Phasor (8)

Relationship between differential, integral operation in phasor listed as follow:

$$v(t) \longleftrightarrow V = V \angle \phi$$

$$\frac{dv}{dt} \longleftrightarrow j\omega V$$

$$\int v dt \longleftrightarrow \frac{V}{j\omega}$$

Using the phasor approach, determine the current $i(t)$ in a circuit described by the integrodifferential equation

$$4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

Solution:

We transform each term in the equation from time domain to phasor domain. Keeping Eqs. (9.27) and (9.28) in mind, we obtain the phasor form of the given equation as

$$4\mathbf{I} + \frac{8\mathbf{I}}{j\omega} - 3j\omega\mathbf{I} = 50\angle 75^\circ$$

But $\omega = 2$, so

$$\mathbf{I}(4 - j4 - j6) = 50\angle 75^\circ$$

$$\mathbf{I} = \frac{50\angle 75^\circ}{4 - j10} = \frac{50\angle 75^\circ}{10.77\angle -68.2^\circ} = 4.642\angle 143.2^\circ \text{ A}$$

Converting this to the time domain,

$$i(t) = 4.642 \cos(2t + 143.2^\circ) \text{ A}$$

Keep in mind that this is only the steady-state solution, and it does not require knowing the initial values.

Three stylized balloons in green, blue, and purple are positioned on the left side of the slide. Each balloon has a string and several small yellow triangular flags attached to it. A blue curved line extends from the middle balloon towards the text.

Phasor Relationships for Circuit Elements (1)

PHASOR RELATIONSHIPS FOR CIRCUIT ELEMENTS

Resistor

For power-line frequencies and frequencies up to a few hundred kilohertz, resistance is, for all practical purposes, unaffected by the frequency of the applied sinusoidal voltage or current. For this frequency region, the resistor R of Fig. 14.4 can be treated as a constant, and Ohm's law can be applied as follows. For $v = V_m \sin \omega t$,

$$i = \frac{v}{R} = \frac{V_m \sin \omega t}{R} = \frac{V_m}{R} \sin \omega t = I_m \sin \omega t$$

where

$$I_m = \frac{V_m}{R}$$

(14.2)

In addition, for a given i ,

$$v = iR = (I_m \sin \omega t)R = I_m R \sin \omega t = V_m \sin \omega t$$

where

$$V_m = I_m R$$

(14.3)

A plot of v and i in Fig. 14.5 reveals that

for a purely resistive element, the voltage across and the current through the element are in phase, with their peak values related by Ohm's law.

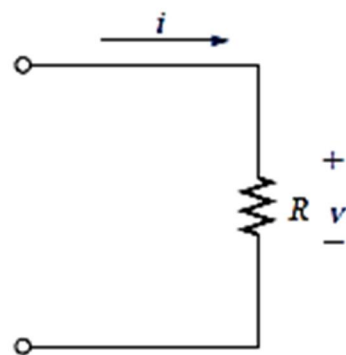


FIG. 14.4

Determining the sinusoidal response for a resistive element.

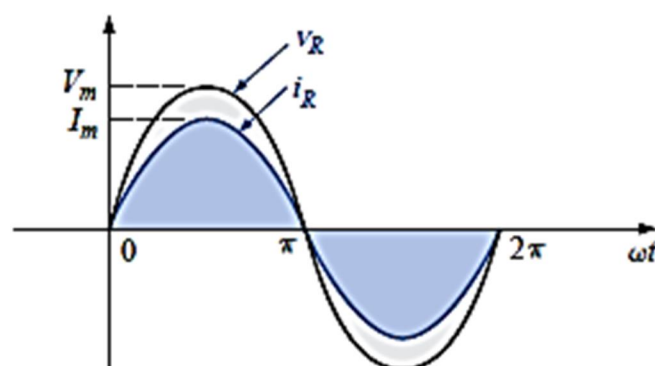


FIG. 14.5

The voltage and current of a resistive element are in phase.

Inductor

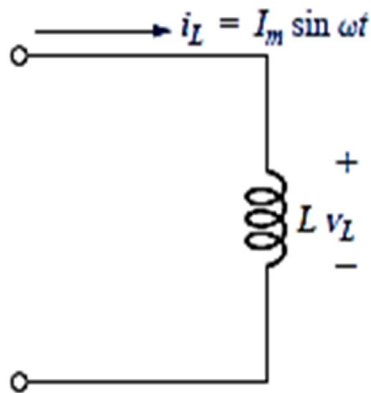


FIG. 14.8

Investigating the sinusoidal response of an inductive element.

$$v_L = L \frac{di_L}{dt}$$

and, applying differentiation,

$$\frac{di_L}{dt} = \frac{d}{dt}(I_m \sin \omega t) = \omega I_m \cos \omega t$$

Therefore,
$$v_L = L \frac{di_L}{dt} = L(\omega I_m \cos \omega t) = \omega L I_m \cos \omega t$$

or
$$v_L = V_m \sin(\omega t + 90^\circ)$$

where
$$V_m = \omega L I_m$$

Note that the peak value of v_L is directly related to ω ($= 2\pi f$) and L as predicted in the discussion above.

A plot of v_L and i_L in Fig. 14.9 reveals that

for an inductor, v_L leads i_L by 90° , or i_L lags v_L by 90° .

Inductance is typified by the behavior of a coil of wire in resisting any change of electric current through the coil and denotes by X_L

$$X_L = \frac{V_m}{I_m} = \frac{\omega L I_m}{I_m} = \omega L$$

Capacitor

$$i_C = C \frac{dv_C}{dt}$$

and, applying differentiation,

$$\frac{dv_C}{dt} = \frac{d}{dt}(V_m \sin \omega t) = \omega V_m \cos \omega t$$

Therefore,

$$i_C = C \frac{dv_C}{dt} = C(\omega V_m \cos \omega t) = \omega C V_m \cos \omega t$$

or

$$i_C = I_m \sin(\omega t + 90^\circ)$$

where

$$I_m = \omega C V_m$$

Note that the peak value of i_C is directly related to ω ($= 2\pi f$) and C , as predicted in the discussion above.

A plot of v_C and i_C in Fig. 14.12 reveals that

*for a capacitor, i_C leads v_C by 90° , or v_C lags i_C by 90° .**

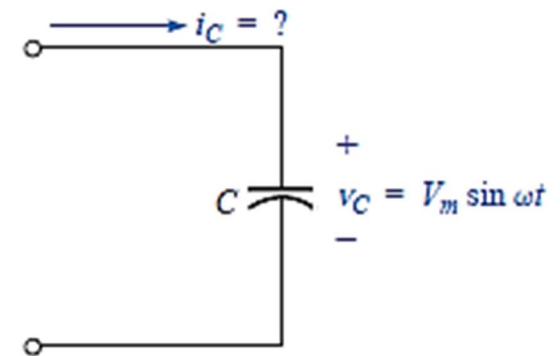


FIG. 14.11

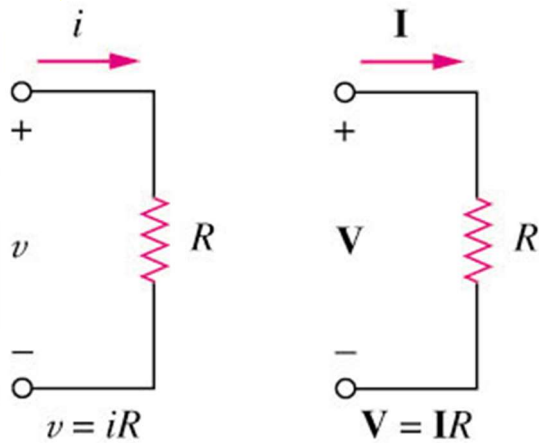
Investigating the sinusoidal response of a capacitive element.

Reactance is the opposition of a circuit element to a change in current or voltage, due to that element's inductance or capacitance. **reactance** of a capacitor is denoted by X_C

$$X_C = \frac{V_m}{I_m} = \frac{V_m}{\omega C V_m} = \frac{1}{\omega C}$$

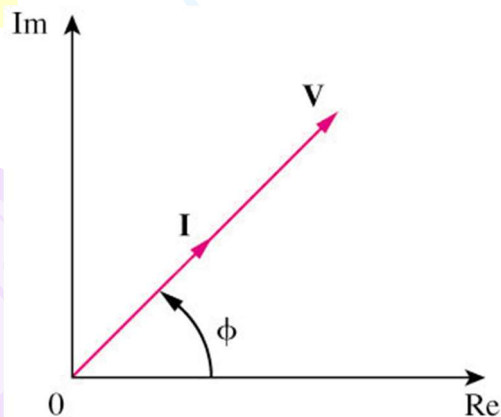
9.4 Phasor Relationships for Circuit Elements (1)

Resistor:

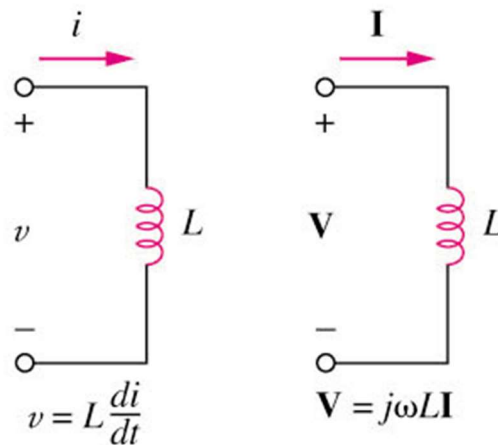


(a)

(b)

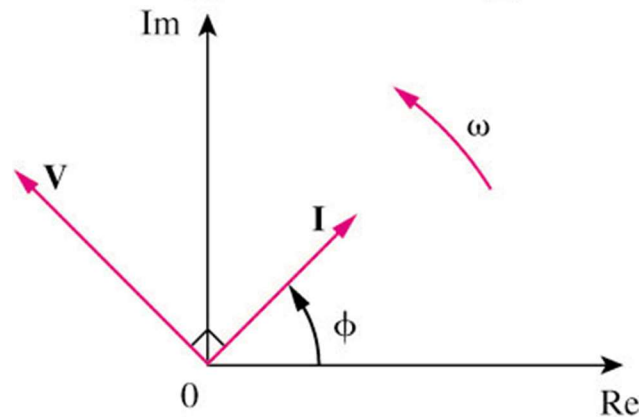


Inductor:

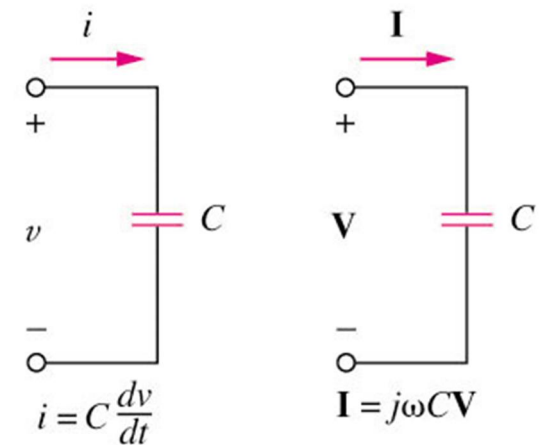


(a)

(b)

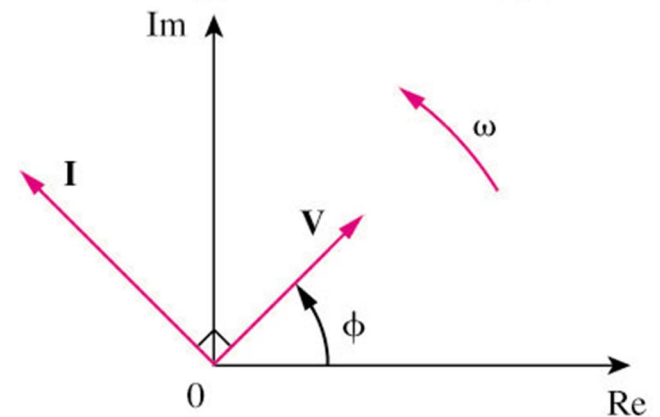


Capacitor:



(a)

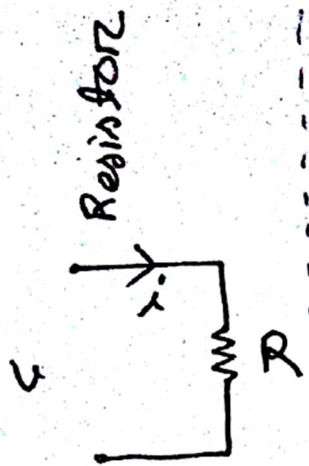
(b)



IMPEDANCE

The impedance **Z** of a circuit is the ratio of the phasor voltage **V** to the phasor current **I**, measured in ohms (Ω).

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} \quad \text{or} \quad \mathbf{V} = \mathbf{Z}\mathbf{I}$$

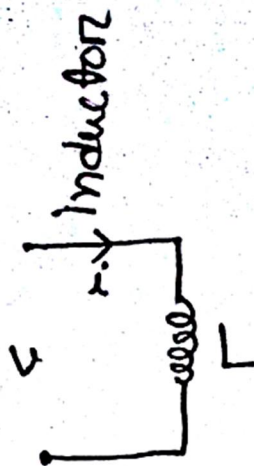


$$V = i R$$

$$V = R I$$

$$Z = \frac{V}{I} = R$$

impedance: R

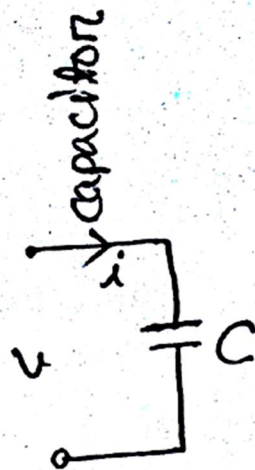


$$V = L \frac{di}{dt}$$

$$V = j\omega L I$$

$$Z = \frac{V}{I} = j\omega L$$

$j\omega L$

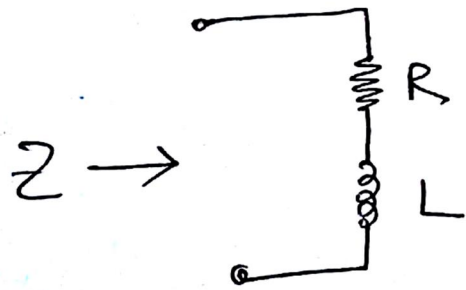


$$i = C \frac{dV}{dt} \quad [\text{time-domain}]$$

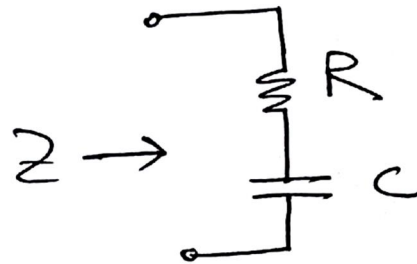
$$I = j\omega C V \quad [\text{frequency-domain}]$$

$$Z = \frac{V}{I} = \frac{1}{j\omega C} = -j \frac{1}{\omega C}$$

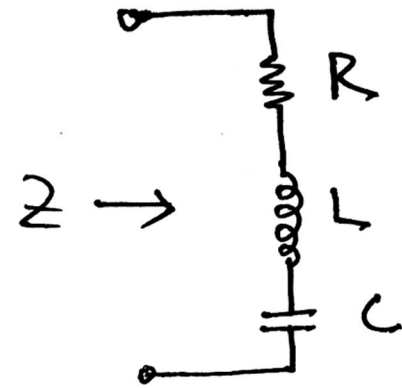
$-j \frac{1}{\omega C}$



$$Z = R + j\omega L$$



$$Z = R - j\frac{1}{\omega C}$$



$$\begin{aligned} Z &= R + j\omega L - j\frac{1}{\omega C} \\ &= R + j\left(\omega L - \frac{1}{\omega C}\right) \end{aligned}$$

In general: $z = R + jX$

$$X = \omega L$$

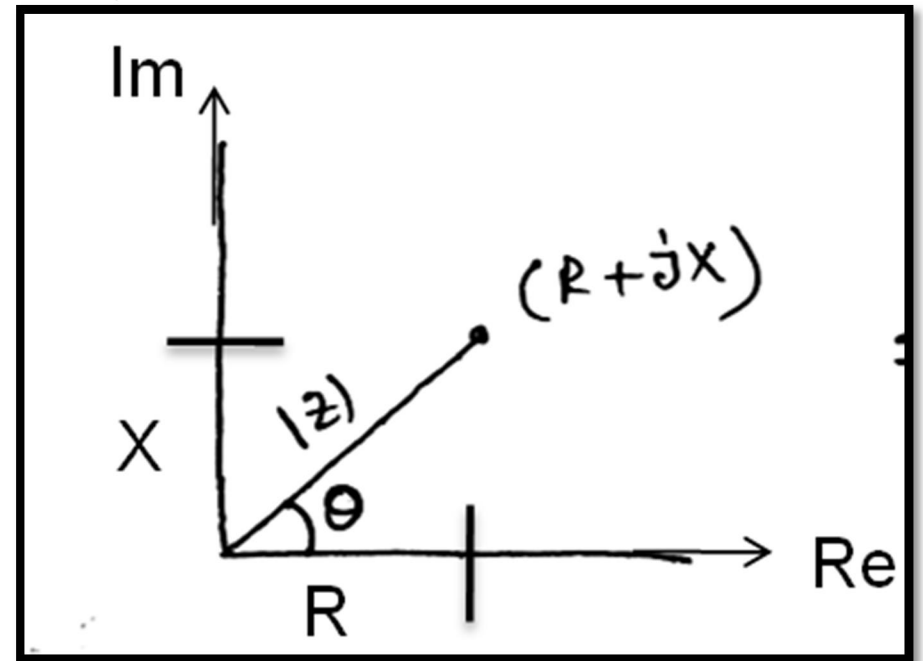
$$X = -\frac{1}{\omega C}$$

$$X = \left(\omega L - \frac{1}{\omega C}\right)$$

in polar form: $z = |z| \angle \theta$

$$|z| = \sqrt{R^2 + X^2}$$

$$\theta = \tan^{-1} \frac{X}{R}$$



$$R = |z| \cos \theta$$

$$X = |z| \sin \theta$$

Find $v(t)$ and $i(t)$ in the circuit shown in Fig.

Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$

Hence the current

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A} \end{aligned} \quad (9.9.1)$$

The voltage across the capacitor is

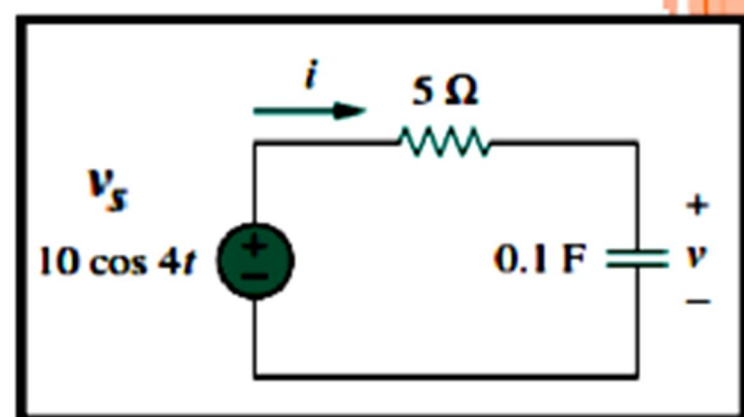
$$\begin{aligned} \mathbf{V} &= \mathbf{I} \mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V} \end{aligned}$$

Converting $\mathbf{I}$ and $\mathbf{V}$ in Eqs. (9.9.1) and (9.9.2) to the time domain, we get

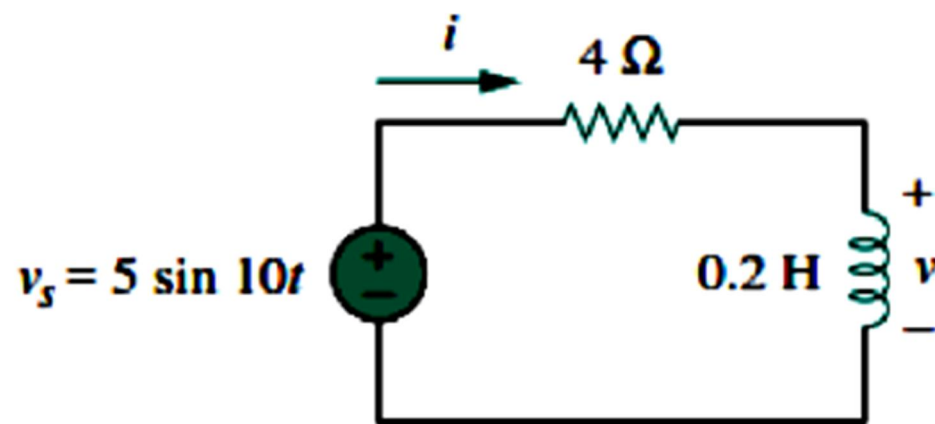
$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.



Determine $v(t)$ and $i(t)$.



Answer: $2.236 \sin(10t + 63.43^\circ) \text{ V}$, $1.118 \sin(10t - 26.57^\circ) \text{ A}$.

9.4 Phasor Relationships for Circuit Elements (2)

Summary of voltage-current relationship

Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$

EXAMPLE 14.1 The voltage across a resistor is indicated. Find the sinusoidal expression for the current if the resistor is $10\ \Omega$. Sketch the curves for v and i .

a. $v = 100 \sin 377t$

b. $v = 25 \sin(377t + 60^\circ)$

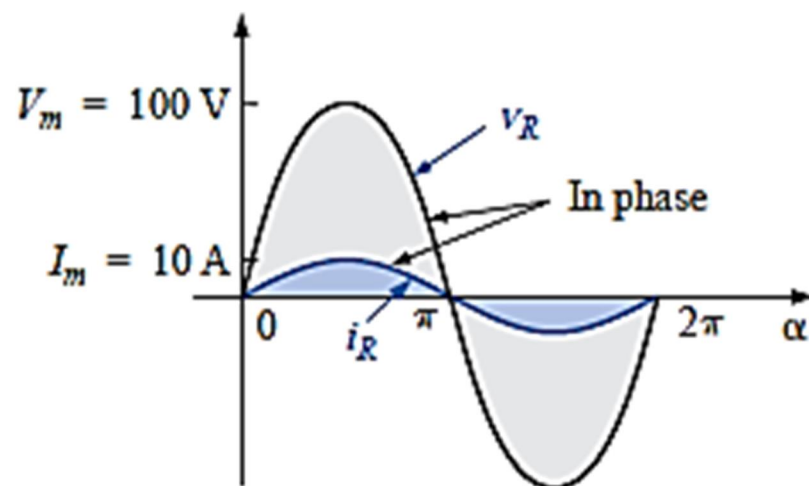
Solutions:

a. Eq. (14.2): $I_m = \frac{V_m}{R} = \frac{100\text{ V}}{10\ \Omega} = 10\text{ A}$

(v and i are in phase), resulting in

$$i = 10 \sin 377t$$

The curves are sketched in Fig. 14.13.



EXAMPLE 14.3 The current through a 0.1-H coil is provided. Find the sinusoidal expression for the voltage across the coil. Sketch the v and i curves.

a. $i = 10 \sin 377t$

b. $i = 7 \sin(377t - 70^\circ)$

Solutions:

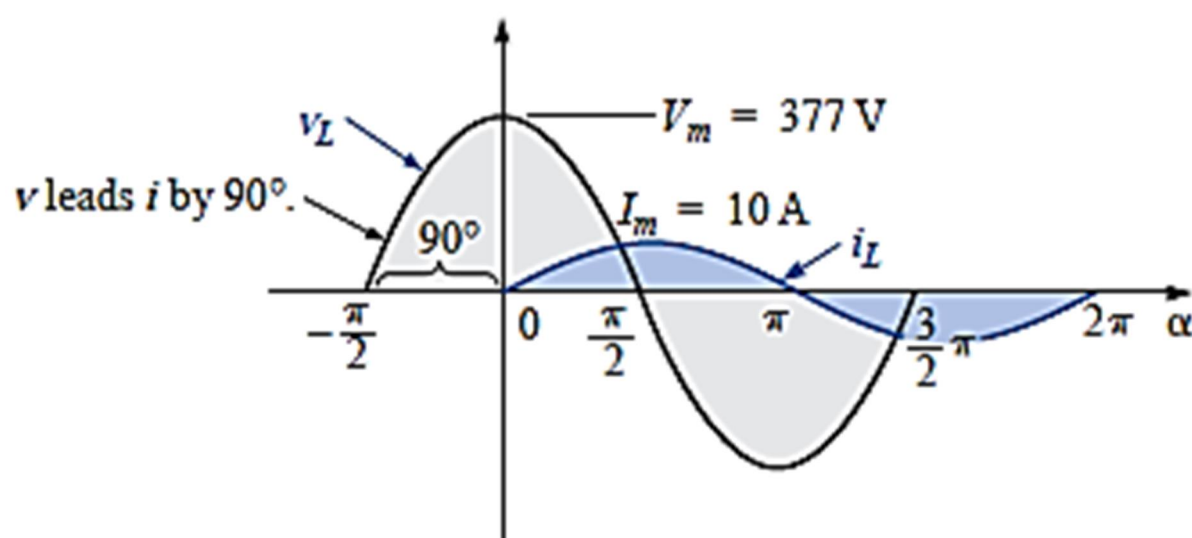
a. Eq. (14.4): $X_L = \omega L = (377 \text{ rad/s})(0.1 \text{ H}) = 37.7 \Omega$

Eq. (14.5): $V_m = I_m X_L = (10 \text{ A})(37.7 \Omega) = 377 \text{ V}$

and we know that for a coil v leads i by 90° . Therefore,

$$v = 377 \sin(377t + 90^\circ)$$

The curves are sketched in Fig. 14.15.



b. X_L remains at 37.7Ω .

$$V_m = I_m X_L = (7 \text{ A})(37.7 \Omega) = 263.9 \text{ V}$$

and we know that for a coil v leads i by 90° . Therefore,

$$v = 263.9 \sin(377t - 70^\circ + 90^\circ)$$

and

$$v = 263.9 \sin(377t + 20^\circ)$$

The curves are sketched in Fig. 14.16.

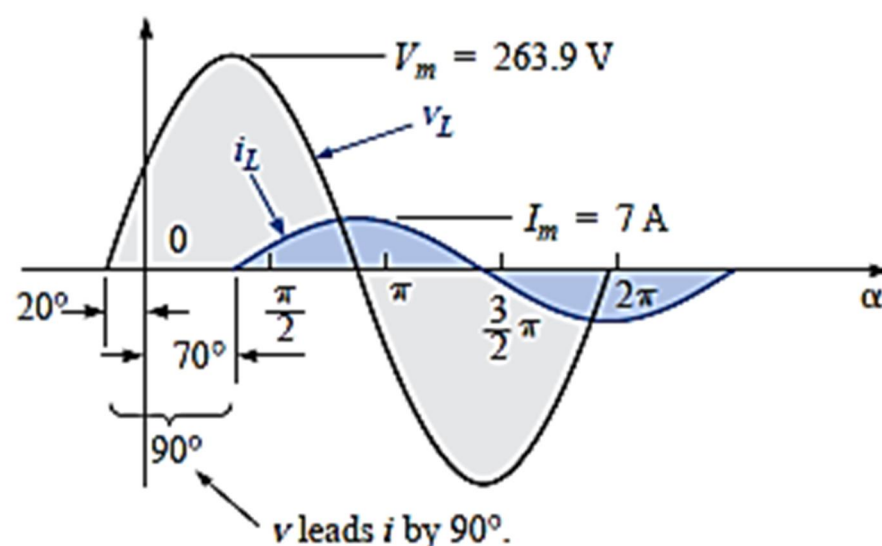


FIG. 14.16
Example 14.3(b).

EXAMPLE 14.5 The voltage across a $1\text{-}\mu\text{F}$ capacitor is provided below. What is the sinusoidal expression for the current? Sketch the v and i curves.

$$v = 30 \sin 400t$$

Solution:

$$\text{Eq. (14.6): } X_C = \frac{1}{\omega C} = \frac{1}{(400 \text{ rad/s})(1 \times 10^{-6} \text{ F})} = \frac{10^6 \Omega}{400} = 2500 \Omega$$

$$\text{Eq. (14.7): } I_m = \frac{V_m}{X_C} = \frac{30 \text{ V}}{2500 \Omega} = 0.0120 \text{ A} = 12 \text{ mA}$$

and we know that for a capacitor i leads v by 90° . Therefore,

$$i = 12 \times 10^{-3} \sin(400t + 90^\circ)$$

The curves are sketched in Fig. 14.17.

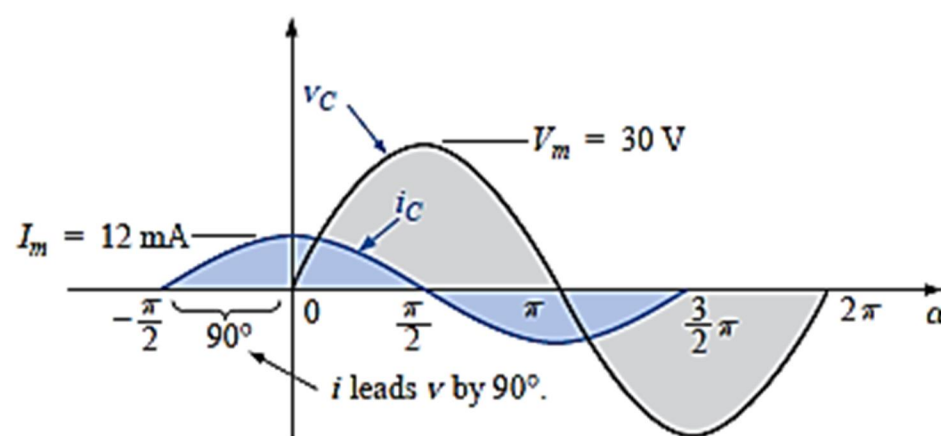


FIG. 14.17
Example 14.5.

EXAMPLE 14.7 For the following pairs of voltages and currents, determine whether the element involved is a capacitor, an inductor, or a resistor, and determine the value of C , L , or R if sufficient data are provided (Fig. 14.18):

- a. $v = 100 \sin(\omega t + 40^\circ)$
 $i = 20 \sin(\omega t + 40^\circ)$
- b. $v = 1000 \sin(377t + 10^\circ)$
 $i = 5 \sin(377t - 80^\circ)$
- c. $v = 500 \sin(157t + 30^\circ)$
 $i = 1 \sin(157t + 120^\circ)$
- d. $v = 50 \cos(\omega t + 20^\circ)$
 $i = 5 \sin(\omega t + 110^\circ)$

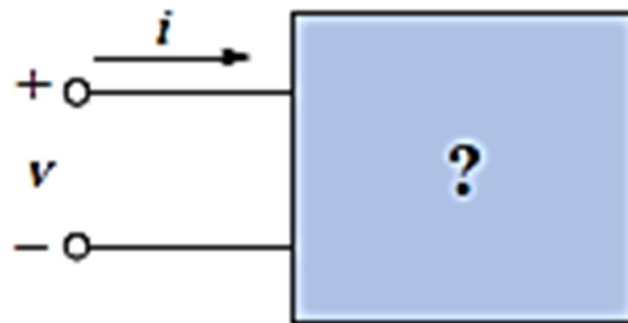


FIG. 14.18
Example 14.7.

9.4 Phasor Relationships for Circuit Elements (3)

Example 7

If voltage $v(t) = 6\cos(100t - 30^\circ)$ is applied to a $50\ \mu\text{F}$ capacitor, calculate the current, $i(t)$, through the capacitor.

Answer: $i(t) = \underline{30\ \cos(100t + 60^\circ)\ \text{mA}}$

9.5 Impedance and Admittance (1)

- The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I , measured in ohms Ω .

$$Z = \frac{V}{I} = R + jX$$

where $R = \text{Re}, Z$ is the resistance and $X = \text{Im}, Z$ is the reactance. **Positive X is for L and negative X is for C.**

- The admittance Y is the reciprocal of impedance, measured in siemens (S).

$$Y = \frac{1}{Z} = \frac{I}{V}$$

9.5 Impedance and Admittance (2)

Impedances and admittances of passive elements

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

Find $v(t)$ and $i(t)$ in the circuit shown in Fig. 9.16.

Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$

Hence the current

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A} \end{aligned} \quad (9.9.1)$$

The voltage across the capacitor is

$$\begin{aligned} \mathbf{V} &= \mathbf{I}Z_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V} \end{aligned} \quad (9.9.2)$$

Converting $\mathbf{I}$ and $\mathbf{V}$ in Eqs. (9.9.1) and (9.9.2) to the time domain, we get

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.

Example 9.9

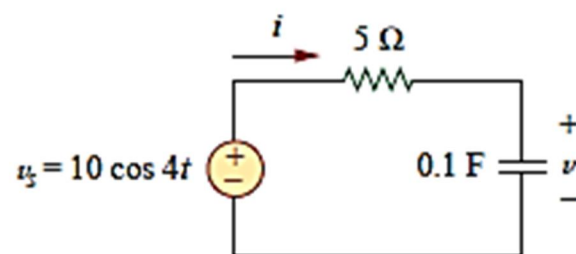


Figure 9.16
For Example 9.9.

Example 9.10

Find the input impedance of the circuit in Fig. 9.23. Assume that the circuit operates at $\omega = 50$ rad/s.

Solution:

Let

Z_1 = Impedance of the 2-mF capacitor

Z_2 = Impedance of the 3- Ω resistor in series with the 10-mF capacitor

Z_3 = Impedance of the 0.2-H inductor in series with the 8- Ω resistor

Then

$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$

$$Z_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \Omega$$

$$Z_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \Omega$$

The input impedance is

$$\begin{aligned} Z_{in} &= Z_1 + Z_2 \parallel Z_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} = -j10 + 3.22 - j1.07 \Omega \end{aligned}$$

Thus,

$$Z_{in} = 3.22 - j11.07 \Omega$$

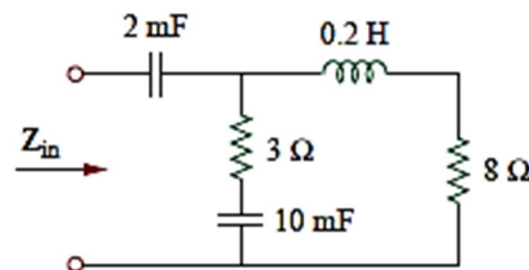


Figure 9.23
For Example 9.10.

Example 9.11

Determine $v_o(t)$ in the circuit of Fig. 9.25.

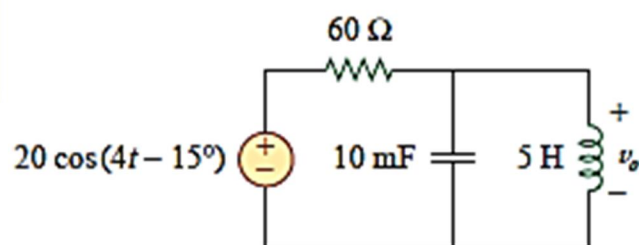


Figure 9.25
For Example 9.11.

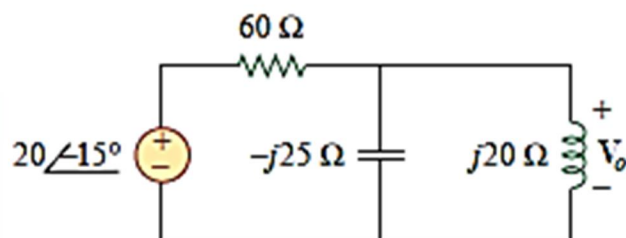


Figure 9.26
The frequency domain equivalent of the circuit in Fig. 9.25.

Solution:

To do the analysis in the frequency domain, we must first transform the time domain circuit in Fig. 9.25 to the phasor domain equivalent in Fig. 9.26. The transformation produces

$$v_s = 20 \cos(4t - 15^\circ) \Rightarrow V_s = 20 \angle -15^\circ \text{ V}, \quad \omega = 4$$

$$10 \text{ mF} \Rightarrow \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} = -j25 \Omega$$

$$5 \text{ H} \Rightarrow j\omega L = j4 \times 5 = j20 \Omega$$

Let

Z_1 = Impedance of the $60\text{-}\Omega$ resistor

Z_2 = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

Then $Z_1 = 60 \Omega$ and

$$Z_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100 \Omega$$

By the voltage-division principle,

$$\begin{aligned} V_o &= \frac{Z_2}{Z_1 + Z_2} V_s = \frac{j100}{60 + j100} (20 \angle -15^\circ) \\ &= (0.8575 \angle 30.96^\circ) (20 \angle -15^\circ) = 17.15 \angle 15.96^\circ \text{ V} \end{aligned}$$

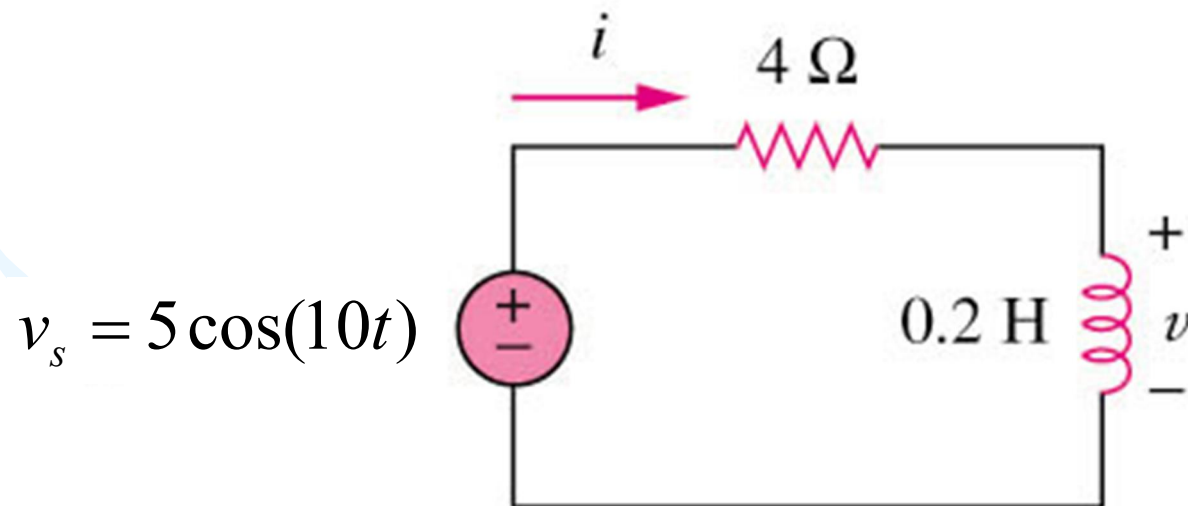
We convert this to the time domain and obtain

$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \text{ V}$$

9.5 Impedance and Admittance (5)

Example 8

Refer to Figure below, determine $v(t)$ and $i(t)$.

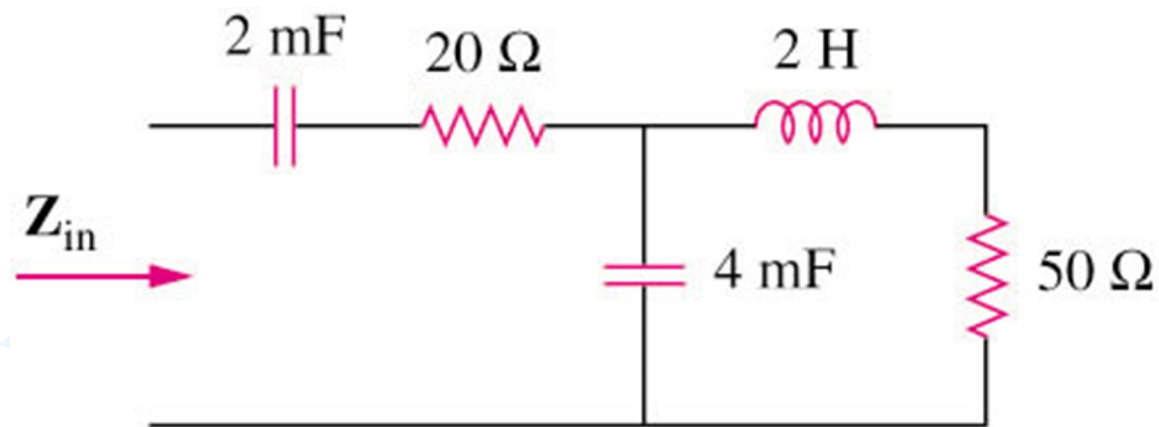


Answers: $i(t) = 1.118 \cos(10t - 26.56^\circ) \text{ A}$; $v(t) = 2.236 \cos(10t + 63.43^\circ) \text{ V}$

9.7 Impedance Combinations (2)

Example 9

Determine the input impedance of the circuit in figure below at $\omega = 10$ rad/s.



Answer: $Z_{in} = 32.38 - j73.76$

Find current I in the circuit of Fig. 9.28.

Example 9.12

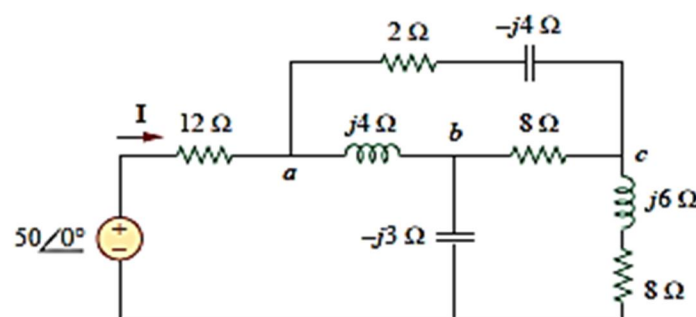


Figure 9.28
For Example 9.12.

Solution:

The delta network connected to nodes a , b , and c can be converted to the Y network of Fig. 9.29. We obtain the Y impedances as follows using Eq. (9.68):

$$Z_{an} = \frac{j4(2 - j4)}{j4 + 2 - j4 + 8} = \frac{4(4 + j2)}{10} = (1.6 + j0.8)\ \Omega$$
$$Z_{bn} = \frac{j4(8)}{10} = j3.2\ \Omega, \quad Z_{cn} = \frac{8(2 - j4)}{10} = (1.6 - j3.2)\ \Omega$$

The total impedance at the source terminals is

$$\begin{aligned} Z &= 12 + Z_{an} + (Z_{bn} - j3) \parallel (Z_{cn} + j6 + 8) \\ &= 12 + 1.6 + j0.8 + (j0.2) \parallel (9.6 + j2.8) \\ &= 13.6 + j0.8 + \frac{j0.2(9.6 + j2.8)}{9.6 + j3} \\ &= 13.6 + j1 = 13.64\angle 4.204^\circ\ \Omega \end{aligned}$$

The desired current is

$$I = \frac{V}{Z} = \frac{50\angle 0^\circ}{13.64\angle 4.204^\circ} = 3.666\angle -4.204^\circ\ \text{A}$$